Contents

0.1 Basic Topology

Problem 0.1.1 ▸ Convex Bodies

A convex body is a compact, convex subset $K \subseteq \mathbb{R}^n$ which contains 0 as an interior point and which is symmetric around the origin .

1. Prove that the following formula defines a norm on $\mathbb{R}^n$:

$$N(x) := \inf\left\{\lambda > 0 : \frac{x}{\lambda} \in K\right\}$$

2. Prove that K is homeomorphic to $\mathbb{D}^n$ and that ∂K is homeomorphicc to $\mathbb{S}^{n-1}$.

Note.

1. 若 K 是一个单位球. 则 $N(x) = \|x\|$.
2. 证明 $N(x)$ 是一个 norm, 需要证明:1. 非负正定 2. 绝对一次齐次 3. 三角不等式.
3. 由于 φ 的形式足够简单, 可以通过直接构造逆映射证明双射.

Lemma 0.1.2

X is compact, Y is Hausdorff . If $\exists f : X \to Y$ is a continious bijection, then X, Y are homeomorphic.

Proof. 1. It is obvious that $N(x) \geq 0$. If $N(x) = 0$, then

$$\inf\left\{\lambda > 0 : \frac{x}{\lambda} \in K\right\} = 0$$

which means that for all $\varepsilon > 0$, $\frac{x}{\varepsilon} \in K$ .Since K is compact, $\operatorname{dist}(K, 0) < \infty$. If $x \neq 0$,then $\|x\| > 0$. We take $\varepsilon_0 < \frac{1}{\operatorname{dist}(K,0)\cdot\|x\|}$. Then $\left\|\frac{x}{\varepsilon_0}\right\| > \operatorname{dist}(K, 0)$. But $\frac{x}{\varepsilon_0} \in K$, which is a contradiction.For the second part, let $x \in \mathbb{R}^n, k \in \mathbb{R} \setminus \{0\}$. For all $\lambda > N(x)$,

$$\frac{x}{\lambda} \in K \implies \frac{kx}{k\lambda} \in K$$

,which shows that $N(kx) \leq kN(x)$. Replace k by $\frac{1}{k}$ and replace x by kx, we have $N(kx) \geq kN(x)$. Thus $N(kx) = kN(x)$. For the triangle inequality , we need to show that

$$\inf\left\{\lambda > 0 \,:\, \frac{x_1 + x_2}{\lambda} \in K\right\} \leq \inf\left\{\lambda > 0 \,:\, \frac{x_1}{\lambda} \in K\right\} + \inf\left\{\lambda > 0 \,:\, \frac{x_2}{\lambda} \in K\right\}$$

Let $\lambda_0 = N(x_1 + x_2)$. Only need to show that

$$\frac{x_1 + x_2}{N(x_1) + N(x_2)} \in K$$

For all $\delta > 0$, we have

$$\frac{x_1}{N(x_1) + \delta} \in K, \quad \frac{x_2}{N(x_2) + \delta} \in K$$

. Since K is convex , we have the convex combination

$$\frac{1}{N(x_1) + N(x_2) + 2\delta}\left((N(x_2) + \delta)\frac{x_2}{N(x_2) + \delta} + (N(x_1) + \delta)\frac{x_1}{N(x_1) + \delta}\right) \in K$$

That is

$$\frac{x_1 + x_2}{N(x_1) + N(x_2) + 2\delta} \in K$$

for all $\delta > 0$. Then

$$N(x_1 + x_2) \leq N(x_1) + N(x_2) + 2\delta, \quad \forall \delta > 0 \implies N(x_1 + x_2) \leq N(x_1) + N(x_2)$$

2. $x \in K^\circ \iff N(x) < 1$, $x \in \operatorname{Ext}(K) \iff N(x) > 1$, $x \in \partial K \iff$

$N(x) = 1$. We define a map $\varphi : \mathbb{R}^n \to \mathbb{R}^n$.

$$\varphi(x) := \begin{cases} N(x)\frac{x}{\|x\|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

We have

$$\|\varphi(x)\| = N(x)$$

Then φ maps K° on to $(\mathbb{D}^n)^\circ$, and maps ∂K on to $\mathbb{S}^{n-1}$. To show φ is continious , only need to show it is continious at 0 . In fact, $\|\lim_{x\to 0} \varphi(x)\| = \lim_{x\to 0} \|\varphi(x)\| = \lim_{x\to 0} N(x) = 0$ since N is a norm . We define $\psi : \mathbb{R}^n \to \mathbb{R}^n$:

$$\psi(x) := \begin{cases} \frac{\|x\|}{N(x)}x, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then $\varphi \circ \psi = \psi \circ \varphi = \mathrm{Id}$. The same skills can be used to show ψ is continious at 0 as well . Finally , by the lemma above, X, Y are homeomorphic.

□

Problem 0.1.3 ▸ Projective spaces

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. The n -dimensional projectinve space $\mathbb{KP}^n$ is the orbit space of $\mathbb{k}^{n+1} \setminus \{0\}$ under the action of $\mathbb{k}^*$ by rescaling.

1. Prove that $\mathbb{RP}^n$ is homeomorphic to the orbit space $\mathbb{S}^n \setminus \{\pm 1\}$, where $\{\pm 1\}$ acts by multiplication on $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$.
2. Prove that $\mathbb{CP}^n$ is homeomorphic to the orbit space $\mathbb{S}^{2n+1} \setminus \mathbb{S}^1$, where $\mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}$ acts on $\mathbb{S}^{2n+1} \subseteq \mathbb{C}^{n+1}$ by multiplication.
3. Prove that $\mathbb{RP}^1$ is homeomorphic to $\mathbb{S}^1$ and that $\mathbb{CP}^1$ is homeomorphic to $\mathbb{S}^2$.

Proof. 1. Define

$$\pi : \mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{RP}^n, \quad \pi(x) = [x]$$

and define

$$f : \mathbb{S}^n \to \mathbb{RP}^n = \pi|_{\mathbb{S}^n}$$

Then f is continuous. Take any $p \in \mathbb{RP}^n$, there exists $v \in \mathbb{R}^{n+1} \setminus \{0\}$, such that $p = [v]$. Let $x = \frac{v}{\|v\|}$, then $x \in \mathbb{S}^n$,

$$f(x) = [x] = [v] = p$$

thus f is surjective. If

$$f(x) = f(y)$$

, then $[x] = [y]$, there exists $\lambda \in \mathbb{R}^*$, such taht $x = \lambda y$. Since

$$\|x\| = \|\lambda y\| = |\lambda| \|y\| \implies |\lambda| = 1$$

, $\lambda = \pm 1$, $x = y$ or $x = -y$.Thus $\text{Fiber}_p(f) = \text{Orbit}_x(\{\pm 1\})$ The universal property of quotient topology gives a continuous bijection

$$\tilde{f} : \mathbb{S}^n \setminus \{\pm 1\} \to \mathbb{RP}^n$$

Since $\mathbb{S}^n \setminus \{\pm 1\}$ is compact, $\mathbb{RP}^n$ is Hausdorff, $\tilde{f}$ is a homeomorphism.

2. Similarly, let $g : \mathbb{S}^{2n+1} \to \mathbb{CP}^n$, $g(z) = [z]$. Take any $p = [w] \in \mathbb{CP}^n$,where $w \in \mathbb{C}^{n+1} \setminus \{0\}$. Let $z = \frac{w}{\|w\|}$. Then

$$g(z) = [z] = [w] = p$$

g is surjective. If

$$g(z_1) = g(z_2)$$

, then there exists $\lambda \in \mathbb{C}^*$, such that $z_1 = \lambda z_2$, then

$$\|z_1\| = |\lambda| \|z_2\| \implies |\lambda| = 1$$

which gives that $\lambda \in \mathbb{S}^1$. Thus $\text{Fiber}_p(f) \simeq \text{Orbit}_x\left(\mathbb{S}^1\right)$. Thus

$$\mathbb{CP}^n \simeq \mathbb{S}^{2n+1}/\mathbb{S}^1$$

3. (a) Only need to shwo that $\mathbb{S}^1/\{\pm 1\} \simeq \mathbb{S}^1$,

$$h(z) = z^2$$

induces a continuous bijection $\tilde{h} : \mathbb{S}^1 \setminus \{\pm 1\} \to \mathbb{S}^1$. $\mathbb{S}^1 \setminus \{\pm 1\}$ is compact, $\mathbb{S}^1$ is Hausdorff, $\tilde{h}$ is a homemophism.

   (b) Since $\mathbb{S}^3/\mathbb{S}^1 \simeq \mathbb{S}^2$.

$\square$

Problem 0.1.4

The torus $\mathbb{T}$ is the quotient of $[0,1]^2$ by the equivalence relation generated by $(x,0)\sim(x,1)$ and $(0,y)\sim(1,y)$ for all $x,y\in[0,1]^2$. Prove that $\mathbb{T}$ is homeomorphic to:

1. The product $\mathbb{S}^1\times\mathbb{S}^1$.
2. The quotient of $\mathbb{R}^2$ under the action of the (discrete) group $\mathbb{Z}^2$ by translations.

Proof.

1. Let
$$f(x,y) := \left(e^{2\pi xi}, e^{2\pi yi}\right)$$
Then
$$f(x,0) = \left(e^{2\pi ix}, 1\right) = f(x,1)$$
$$f(0,y) = \left(1, e^{2\pi yi}\right) = f(1,y)$$
Thus f induces a map $\bar{f}$ on $\mathbb{T}$ such that $\bar{f}([(x,y)]) = \left(e^{2\pi xi}, e^{2\pi yi}\right)$, which is a continuous function. It is not hard to see that $\bar{f}:\mathbb{T}\to\mathbb{S}^1\times\mathbb{S}^1$ is a bijection. As a continuous image of $[0,1]^2$, $\mathbb{T}$ is compact. Take any closed subset V of $\mathbb{T}$, V is compact. $\bar{f}(V)$ is compact as the continuous image of V. Since $\mathbb{S}^1\times\mathbb{S}^1$ is Hausdorff, $\bar{f}(V)$ is closed. Thus $\bar{f}$ is a continuous and closed bijection, thus is a homeomorphism.
2. Let
$$f(x,y) = (x,y)/\mathbb{Z}^2$$
Then Since $(x,0)-(x,1)\in\mathbb{Z}^2, (0,y)-(1,y)\in\mathbb{Z}^2$. f induces a continuous map $\bar{f}$ on from $\mathbb{T}$ to $\mathbb{S}^1\times\mathbb{S}^1$. It is clear that f maps onto $\mathbb{R}^2/\mathbb{Z}^2$. If $f(x_1,y_1) = f(x_2,y_2)$, then
$$(x_1-x_2, y_1-y_2)\in\mathbb{Z}^2 \implies x_1 = x_2 \text{ or } \{x_1,x_2\} = \{0,1\}, \text{and } y_1 = y_2 \text{ or } \{y_1,y_2\} = \{$$
thus $(x_1,y_1)\sim(x_2,y_2)$. Again since $\mathbb{T}$ is compact, $\mathbb{R}^2/\mathbb{Z}^2$ is Hausdorff, $\bar{f}$ is a homemorphism.

□

Problem 0.1.5

Consider the orthogonal group $O_{n-1}(\mathbb{R})$ as the subgroup of $O_n(\mathbb{R})$ of matrices of the form:

$$\begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix} \in O_n(\mathbb{R}), A \in O_{n-1}(\mathbb{R}).$$

Let $O_{n-1}(\mathbb{R})$ act on $O_n(\mathbb{R})$ by left multiplication. Prove that the orbit space $O_n(\mathbb{R})/O_{n-1}(\mathbb{R})$ is homeomorphic to $\mathbb{S}^{n-1}$.

Proof sketch.

将 $O_n(\mathbb{R})$ 刻画为 $\mathbb{S}^{n-1}$ 上的群作用. 此时 $A \sim \begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix}$ 就是北极点的稳定子群. 一但作用是传递的, 由流形上的轨道-稳定化子定理, 就有

$$O_n(\mathbb{R}) / O_{n-1}(\mathbb{R}) \simeq \mathbb{S}^{n-1}$$

0.2 Homotopy

Problem 0.2.1 ▸ Mobius band

Let M be the Möbius band, that is, the quotient of $[0,1]^2$ under the equivalence relation generated by $(x, 0) \sim (1-x, 1)$ for all $x \in [0,1]$. Prove that M is homotopy equivalent to $\mathbb{S}^1$.

Proof. Define a map

$$H : [0,1]^2 \times [0,1] \to \mathbb{S}^1, \quad H((x,y),t) = \left((1-t)x + \frac{1}{2}t, y\right)$$

Let $\tilde{H} = \pi \circ H$. Then

$$\tilde{H}((x,0),t) = \pi\left((1-t)x + \frac{1}{2}t, 0\right)$$

$$\tilde{H}((1-x,1),t) = \pi\left(\left((1-t)(1-x) + \frac{1}{2}t\right), 1\right)$$

Since

$$1 - \left((1-t)x + \frac{1}{2}t\right) = (1-t)(1-x) + \frac{1}{2}t$$

,

$$\tilde{H}((x,0),t) = \tilde{H}((1-x,1),t)$$

$\tilde{H}$ induces a continuous map H^* on M. And

$$H^*([(x,y),0]) = [(x,y)], \quad H^*([(x,y),1]) = \left[\left(\frac{1}{2},y\right)\right]$$

Thus H^* is a deformation restract frome M to $\mathbb{S}^1$, which is a homotopy equivalence as well. □

Problem 0.2.2

Let X be space. The cone CX is the quotient space:

$$CX = (X \times [0,1])/(X \times \{0\}).$$

Let $\pi : X \times [0,1] \to CX$ be the quotient map and $\iota : X \to CX, x \mapsto \pi(x,1)$.

1. Let $f : X \to Y$ be a map. Prove that f is homotopic to a constant map if and only if there exists $f' : CX \to Y$ such that $f' \circ \iota = f$.
2. Prove that CS^n is homeomorphic to D^{n+1}.
3. Prove that for all X, CX is contractible.

Proof. 1. If such f' exists. Then $H := f' \circ \pi : X \times [0,1] \to Y$ is a homotopy from f to the constant map valued the vetex of CX. Conversly, if f is homotopic to constant map. Suppose $H : X \times I \to Y$ is its homotopy map. Since CX is the quotient space of $X \times [0,1]$, and $H(X \times \{0\}) =$ constant. It follows frome the universal property that H induces a map $f' : CX \to Y$ such that $f' \circ \pi = H$. Finally

$$(f' \circ \iota)(x) = f'(\pi(x,1)) = H(x,1) = f(x)$$

2. Define
$$g : \mathbb{S}^n \times [0,1] \to D^{n+1}, \quad g(x,t) = t \cdot x$$
Since $g(x,0) \equiv 0 \in D^{n+1}$. g induces a continuous function $h : CS^n \to D^{n+1}$ such that $g \circ \pi = h$. We now show that h is a bijection. If
$$h([x_1,t_1]) = h([x_2,t_2])$$
, that is $t_1x_1 = t_2x_2$, then
$$\|t_1x_1\| = \|t_2x_2\| \implies t_1 = t_2$$
If $t_1 = t_2 = 0$, then $[x_1,t_1] = [x_2,t_2] = 0$. If $t_1 = t_2 > 0$, then $x_1 = x_2$, thus $[x_1,t_1] = [\times 2,t_2]$. Thus h is injective. For surjectivity, let $y \in D^{n+1}$.

If $y = 0$, we take $t = 0$, then $h([x, 0] = 0) = y$. If $y \neq 0$, then

$$h\left(\left[\frac{y}{\|y\|}, \|y\|\right]\right) = y$$

Thus h is surjective. Since CS^n is compact, D^{n+1} is Hausdorff, h is homemorphism.

3. Define

$$H : CX \times [0, 1] \to CX, \quad H([x, s], t) = [x, s \cdot (1 - t)]$$

Then it is not hard to see that H is well-defined and continuous. Since

$$H([x, s], 0) = [x, s], \quad H([x, s], 1) = [x, 0]$$

H is a homotopy from Id to Constant .

$\square$

Problem 0.2.3 ▸ Linear Groups

1. Prove that the inclusion $O_n(\mathbb{R}) \to GL_n(\mathbb{R})$ is a homotopy equivalence. (Use the Gram-Schmidt orthonormalization procedure to construct the reverse map).
2. Among the following matrix groups, determine which ones are compact, and determine their π_0: $GL_n(\mathbb{C})$, $GL_n(\mathbb{R})$, $O_n(\mathbb{R})$, $SO_n(\mathbb{R})$, $U_n(\mathbb{C})$, $SU_n(\mathbb{C})$.

Proof. 1. Take $A \in \mathrm{GL}_n(\mathbb{R})$, suppose $A = (v_1, \cdots, v_n)$. Then $\{v_1, \cdots, v_n\}$ passing to an orthonormal basis $\{u_1, \cdots, u_n\}$ by Gram-Schmidt orthonormalization. We write $r(A) = \{u_1, \cdots, u_n\}$. Then r is a continuous map on $\mathrm{GL}_n(\mathbb{R})$ since the inner product is continuous. We define

$$H : \mathrm{GL}_n(\mathbb{R}) \times [0, 1] \to \mathrm{GL}_n(\mathbb{R}), \quad H(A, t) = (1 - t)A + tr(A)$$

we need to verify that $H(A, t) \in \mathrm{GL}_n(\mathbb{R})$ for all A, t . The k-th column vector of $H(A, t)$ is $(1 - t)v_k + tu_k$. Since

$$v_k \in \mathrm{span}\{u_1, \cdots, u_{k-1}\}$$

we can write

$$(1-t)\,v_k + tu_k = (1-t)\sum_{j=1}^{k} c_{kj}u_j + tu_k$$

Since $c_{jj} > 0$, $(1-t)\,c_{jj} + t > 0$, $\{(1-t)\,v_1 + tu_1, \cdots, (1-t)\,v_k + tu_k\}$ are linealy independent. And since

$$H\,(A,0) = A = \mathrm{Id}_{\mathrm{GL}_n(\mathbb{R})}\,(A)$$

$$H\,(A,1) = r\,(A) \in O_n\,(\mathbb{R})$$

and for $B \in O_n\,(R)$,$H\,(B,t) = r\,(B) = 1$. We have $H\,:\,\mathrm{Id}_{\mathrm{GL}_n(\mathbb{R})} \simeq i$ is a SDR.

2. $\mathrm{GL}_n\,(\mathbb{C})$ is a open submanifold of $M_n\,(\mathbb{C})$, which is noncompact. Since $\mathrm{GL}_n\,(\mathbb{C})$ is path-connected. $\pi_0\,(\mathrm{GL}_n\,(\mathbb{C})) = 1$. $\mathrm{GL}_n\,(\mathbb{R})$ is a open submanifold of $M_n\,(\mathbb{R})$, thus $Gl_n\,(\mathbb{R})$ is noncompact. $\det\,:\,\mathrm{GL}_n\,(\mathbb{R}) \to \mathbb{R}\setminus\{0\}$ is a continuous surjection. Since $\mathbb{R}\setminus\{0\}$ has two components. $\mathrm{GL}_n\,(\mathbb{R})$ has at least two components. Since

$$\mathrm{GL}_n^+\,(\mathbb{R}) = \{A\,:\,\det A > 0\}\,,\quad \mathrm{GL}_n^-\,(\mathbb{R}) = \{A\,:\,\det A < 0\}$$

are exactly two components of $\mathrm{GL}_n\,(\mathbb{R})$. $\pi_0\,(\mathrm{GL}_n\,(\mathbb{R})) = 2$. Since $\left\{A\,:\,A^TA = E\right\}$ is closed. $O_n\,(\mathbb{R})$ is bounded. $O_n\,(\mathbb{R})$ is compact. $\pi_0\,(O_n\,(\mathbb{R})) = 2$. $SO_n\,(\mathbb{R})$ is a closed subset of $O_n\,(\mathbb{R})$, thus is compact as well. $\pi_0\,(SO_n\,(\mathbb{R})) = 1$. $U_n\,(\mathbb{C})$ is a closed subset of $\mathrm{GL}_n\,(\mathbb{C})$. $\pi_0\,(U_n\,(\mathbb{C})) = 1$. $SU_n\,(\mathbb{C})$ is compact, $\pi_0\,(SU_n\,(\mathbb{C})) = 1$.

□

Problem 0.2.4

1. Prove that $\mathbb{CP}^{n+1}$ is obtained from $\mathbb{CP}^n$ by gluing a cell of dimension $2n+2$.
2. Compute the fundamental group of $\mathbb{CP}^n$ for $n \geq 1$.

0.3 van Kampen

Lemma 0.3.1

设 X 分解为一些道路连通集 A_α 的并, 且每个 A_α 包含了基点 $x_0 \in X$. 则所有含入映射 $A_\alpha \hookrightarrow X$ 的诱导同态 $j_\alpha : \pi_1(A_\alpha) \to \pi_1(X)$ 扩张为同态 $\Phi : *_\alpha \pi_1(A_\alpha) \to \pi_1(X)$.

Proposition 0.3.2

令 $i_{\alpha\beta} : \pi_1(A_\alpha \cap A_\beta) \to \pi_1(A_\alpha)$ 表示含入映射 $A_\alpha \cap A_\beta \hookrightarrow A_\alpha$ 诱导的同态, 则 $j_\alpha i_{\alpha\beta} = j_\beta i_{\beta\alpha}$. 所以 $\ker \Phi$ 包含所有形如 $i_{\alpha\beta}(\omega) i_{\beta\alpha}(\omega)^{-1}, \omega \in \pi_1(A_\alpha \cap A_\beta)$ 的元素.

Note.
即 Φ 将 $A_\alpha \cap A_\beta$ 落在 A_α 和 A_β 中部分的像等同.

Theorem 0.3.3 ▸ van Kampen

设 X 分解为包含了基点 $x_0 \in X$ 的道路连通开集 A_α 的并. 且每个交集 $A_\alpha \cap A_\beta$ 是道路连通的, 则同态 $\Phi : *_\alpha \pi_1(A_\alpha) \to \pi_1(X)$ 是满射. 此外, 若每个 $A_\alpha \cap A_\beta \cap A_\gamma$ 也是道路连通的, 则 Φ 的核 N 由全体形如 $i_{\alpha\beta}(\omega) i_{\beta\alpha}(\omega)^{-1}, \omega \in \pi_1(A_\alpha \cap A_\beta)$ 的元素生成, 从而 Φ 诱导出同构

$$\pi_1(X) \simeq *_\alpha \pi_1(A_\alpha)/N = *_\alpha \pi_1(A_\alpha)/\left(i_{\alpha\beta}(\omega) = i_{\beta\alpha}(\omega)\right)$$

Example 0.3.4

设 $\{X_\alpha\}$ 是一族道路连通的空间, $x_\alpha \in X_\alpha$ 是基点, 考虑它们的楔和 $\bigvee_\alpha X_\alpha$. 对于每个 x_α , 它在 X_α 中都存在形变收缩到自身的开邻域 U_α. 令 $A_\alpha = X_\alpha \bigvee_{\beta \neq \alpha} U_\beta$, 则 A_α 形变收缩到 X_α . 此外, $\bigcap_\alpha A_\alpha$ 形变收缩到点 x_α. 由 van Kampen 定理, $\Phi : \pi_1(X) \to *_\alpha \pi_1(A_\alpha) \simeq *_\alpha \pi_1(X_\alpha)$ 是同构.

0.4 Homology

Lemma 0.4.1 ▸ Snake lemma

Given a commutative diagram of R-module homomorphisms: where the two horizontal sequences are exact, there exists a R-module homomorphism

$$\begin{tikzcd}
 & M' \arrow[r, "\alpha"] \arrow[d, "f'"] & M \arrow[r, "\beta"] \arrow[d, "f"] & M'' \arrow[r] \arrow[d, "f''"] & 0 \\
0 \arrow[r] & N' \arrow[r, "\alpha'"] & N \arrow[r, "\beta'"] & N'' &
\end{tikzcd}$$

$\delta : \operatorname{Ker} f'' \longrightarrow \operatorname{Coker} f'$, called the *connecting homomorphism* such that the sequence

$$\operatorname{Ker} f' \xrightarrow{\bar{\alpha}} \operatorname{Ker} f \xrightarrow{\bar{\beta}} \operatorname{Ker} f'' \xrightarrow{\delta} \operatorname{Coker} f' \xrightarrow{\bar{\alpha}'} \operatorname{Coker} f \xrightarrow{\bar{\beta}'} \operatorname{Coker} f''$$

is exact. Moreover, the ***connecting homomorphism*** δ has the naturality properties, so that the above assignment of a 'snake' to the corresponding 'six-term' exact sequence of modules defines a covariant functor.

Note.
最关键的信息是 ***connetcting homomorphism*** δ. 构造的过程就是借助 β 的是 surjective, α' 是 injectinve, 来调整使得 β 和 α' 对应的箭头是可以反转的. 事实上在应用时, 这个 connetcting homomorphism 的具体构造往往比它的存在性更重要.

Proof. ***to be finished*** We need to show

1. $\bar{\alpha}, \bar{\beta}$ are well defined, where $\bar{\alpha} := \alpha|_{\ker f'}, \bar{\beta} := \beta|_{\ker f}$
2. $\operatorname{Im} \bar{\alpha} = \ker \bar{\beta}$.
3. $\bar{\alpha}', \bar{\beta}'$ are well defined, where $\bar{\alpha}'([x]) := [\alpha'(x)]\,, \bar{\beta}'([y]) := [\beta'(y)]$
4. $\operatorname{Im} \bar{\alpha}' = \ker \bar{\beta}'$
5. Construct a δ.
6. $\operatorname{Im} \bar{\beta} = \ker \delta$
7. $\operatorname{Im} \delta = \ker \bar{\alpha}'$

We prove the above points below.

1. By commutative, $f \circ \alpha (\ker f') = \alpha' \circ f' (\ker f') = 0$, which shows that $\alpha (\ker f') \subseteq \ker f$. Thus $\bar{\alpha}$ is well defined. $\bar{\beta}$'s is similar.
2. $$\operatorname{Im} \bar{\alpha} = \alpha (\ker f') \subseteq \operatorname{Im} \alpha = \ker \beta \implies \beta (\operatorname{Im} \bar{\alpha}) = 0 \implies \operatorname{Im} \bar{\alpha} \subseteq \ker \bar{\beta}$$

   The other side,

   $$\ker \bar{\beta} = \ker \beta \cap \ker f = \operatorname{Im} \alpha \cap \ker f$$

   Take $y \in \ker \bar{\beta}$, there exists $x \in M'$ such that $y = \alpha (x)$ and $f (y) = f (\alpha (x)) = 0$. Since $f \circ \alpha = \alpha' \circ f'$, $\alpha' \circ f' (x) = 0$. Since α' is injective, $x \in \ker f'$. Thus $y \in \operatorname{Im} \bar{\alpha}$.
3. For $\bar{\alpha}'$, we need to show that for $x_1, x_2 \in N'$ such that $x_1 - x_2 \in \operatorname{Im} f'$, $\alpha' (x_1) - \alpha' (x_2) \in \operatorname{Im} f$. It is true by commutative

   $$\alpha' (x_1) - \alpha' (x_2) = \alpha' (x_1 - x_2) \in \operatorname{Im} (\alpha' \circ f') = \operatorname{Im} (f \circ \alpha) \subseteq \operatorname{Im} f$$

   $\bar{\beta}'$'s is similar.
4. Take $[y]_{\sim \operatorname{Im} f} \in \operatorname{Im} \bar{\alpha}'$, then there exists $[x]_{\sim \operatorname{Im} f'} \in \operatorname{Coker} f'$, such that $\alpha' \left([x]_{\sim \operatorname{Im} f'} \right) = [y]_{\sim \operatorname{Im} f}$. Then $\alpha' (x) - y \in \operatorname{Im} f$, there exists $z \in M$, such that

   $$\alpha' (x) - y = f (z)$$

   By commutative and exactness: $\ker \beta' = \operatorname{Im} \alpha'$,

   $$\beta' (y) = \beta' (\alpha' (x) - f (z)) = 0 - \beta' \circ f (z) = -f'' \circ \beta (z) \in \operatorname{Im} f''$$

   , which shows that $\bar{\beta}' \left([y]_{\sim \operatorname{Im} f} \right) = [0]_{\sim \operatorname{Im} f''}$, $\operatorname{Im} \bar{\alpha}' \subseteq \ker \bar{\beta}'$.

   Then, take $[y]_{\sim \operatorname{Im} f} \in \ker \bar{\beta}'$. Then $y \in \ker \beta' = \operatorname{Im} \alpha'$. There exists $x \in N'$ such that $y = \alpha' (x)$. Then $[y]_{\sim \operatorname{Im} f} = \alpha^{\bar{\prime}} \left([x]_{\sim \operatorname{Im} f'} \right)$, $[y]_{\sim \operatorname{Im} f} \in \operatorname{Im} \bar{\alpha}'$, $\ker \bar{\beta}' \subseteq \operatorname{Im} \bar{\alpha}'$.
5. Take $z \in \ker f''$. Since β is surjective, there exists $y \in M$ such that $\beta (y) = z$. Once we show that $f (y) \in \operatorname{Im} \alpha'$, we can define

   $$\delta (z) = [(\alpha')^{-1} (f (y))]_{\sim \operatorname{Im} f'}$$

provided that this definition is well defined. By commutative

$$\beta'(f(y)) = f''(\beta(y)) = f''(z) = 0.$$

Hence $f(y) \in \ker \beta' = \operatorname{Im} \alpha'$. Finally, to show that it is well defined, we take $z_1, z_2 \in \ker f''$ such that $\beta(y_1) = \beta(y_2) = z$. We need to show that $(\alpha')^{-1}(f(y_1 - y_2)) \in \operatorname{Im} f'$. Since $\beta(y_1) - \beta(y_2) = 0$, $\beta(y_1 - y_2) = 0$, $y_1 - y_2 \in \ker \beta = \operatorname{Im} \alpha$. Suppose $\alpha(x) = y_1 - y_2$. Then

$$(\alpha')^{-1}(f(y_1 - y_2)) = (\alpha')^{-1}(f\alpha(x)) = (\alpha')^{-1}(\alpha' \circ f')(x) = f'(x) \in \operatorname{Im} f'$$

, which completes the proof.

□

Corollary 0.4.2

Consider the following commutative diagram of R-modules and R-linear maps in which the two rows are exact. If f_1 and f_3 are isomorphisms then so is f_2.

$$\begin{tikzcd}
0 \arrow[r] & M_1 \arrow[r, "\alpha_1"] \arrow[d, "f_1"] & M_2 \arrow[r, "\alpha_2"] \arrow[d, "f_2"] & M_3 \arrow[r] \arrow[d, "f_3"] & 0 \\
0 \arrow[r] & N_1 \arrow[r, "\beta_1"] & N_2 \arrow[r, "\beta_2"] & N_3 \arrow[r] & 0
\end{tikzcd}$$

Proof. By snake lemma, we have

$$0 \xrightarrow{\alpha_1} \ker f_2 \xrightarrow{\alpha_2} 0 \xrightarrow{\delta} 0 \xrightarrow{\beta_1} \operatorname{Coker} f_2 \xrightarrow{\beta_2} 0$$

is exact. Then we have

$$\begin{aligned} 0 &= \ker \alpha_2 = \operatorname{Im} \alpha_1 = 0 \\ &= \ker f_2 \end{aligned}$$

which shows that $\ker f_2 = 0$. Similarly, $\operatorname{Coker} f_2 = 0$. The above shows that f_2 is an isomorphisim. □

Corollary 0.4.3 ▸ Four lemma

Consider the following commutative diagram of R modules and R-linear maps in which the two rows are exact. Suppose that f_1 is surjective and

f_4 is injective. Then

(i) f_2 is injective $\implies$ f_3 is injective.

(ii) f_3 is surjective $\implies$ f_2 is surjective.

$$\begin{tikzcd} M_1 \arrow[r, "\alpha_1"] \arrow[d, "f_1"] & M_2 \arrow[r, "\alpha_2"] \arrow[d, "f_2"] & M_3 \arrow[r, "\alpha_3"] \arrow[d, "f_3"] & M_4 \arrow[d, "f_4"] \\ N_1 \arrow[r, "\beta_1"] & N_2 \arrow[r, "\beta_2"] & N_3 \arrow[r, "\beta_3"] & N_4 \end{tikzcd}$$

Proof sketch.
如果 f_3 被两个单射 f_2, f_4 夹在中间, 得益于 f_1 是满射, 可以将右三列保持单射性地调整为一条 snake, 从而导出 f_3 是单射.
类似地, 如果 f_2 被两个满射 f_1, f_3 夹在中间, 得益于 f_4 是单射, 左三列可以保持满射性地调整为一条 snake, 导出 f_2 是满射.

Proof. 1. We can construct a snake

$$\begin{tikzcd} & M_2/\ker\alpha_2 \arrow[r, "\alpha_2"] \arrow[d, "\bar{f}_2"] & M_3 \arrow[r, "\alpha_3"] \arrow[d, "f_3"] & \operatorname{Im}\alpha_3 \arrow[r] \arrow[d, "f_4|_{\operatorname{Im}\alpha_3}"] & 0 \\ 0 \arrow[r] & N_2/\ker\beta_2 \arrow[r, "\beta_2"] & N_3 \arrow[r, "\beta_3"] & N_4 & \end{tikzcd}$$

By snake lemma, the following sequence is exact

$$\ker \bar{f}_2 \to \ker f_3 \to \ker f_4|_{\operatorname{Im}\alpha_3}$$

Since f_4 is injective, $f_4|_{\operatorname{Im}_{\alpha_3}}$ is as well. Furthermore, by applying the snake lemma to the following diagram

$$\begin{tikzcd} & M_1 \arrow[r, "\alpha_1"] \arrow[d, "\bar{f}_1"'] & M_2 \arrow[r, "p"] \arrow[d, "f_2"] & M_2/\ker\alpha_2 \arrow[r] \arrow[d, "\bar{f}_2"] & 0 \\ 0 \arrow[r] & N_1/\ker\beta_1 \arrow[r, "\bar{\beta}_1"'] & N_2 \arrow[r, "p"'] & N_2/\ker\beta_2 & \end{tikzcd}$$

We get

$$0 = \ker f_2 \to \ker \bar{f}_2 \xrightarrow{\delta} \operatorname{Coker} \bar{f}_1$$

is a exact suquence,where $\operatorname{Coker} \bar{f}_1 = 0$ since f_1 is surjective and $\bar{f}_1$ is as well. It shows that $\ker \bar{f}_2 = 0$. Finally, we have the following sequence is exact

$$0 = \ker \bar{f}_2 \to \ker f_3 \to \ker f_4|_{\operatorname{Im}\alpha_3} = 0$$

It follows that $\ker f_3 = 0$, f_3 is injective.

2. Another snake we can construct is

$$\begin{tikzcd}
 & M_1 \arrow[r, "\alpha_1"] \arrow[d, "\bar{f}_1"] & M_2 \arrow[r, "\alpha_2"] \arrow[d, "f_2"] & \operatorname{Im}\alpha_2 = \ker\alpha_3 \arrow[r] \arrow[d, "f_3|_{\operatorname{Im}\alpha_2}"] & 0 \\
0 \arrow[r] & N_1/\ker\beta_1 \arrow[r, "\bar{\beta}_1"] & N_2 \arrow[r, "\beta_2"] & \operatorname{Im}\beta_2 = \ker\beta_3 &
\end{tikzcd}$$

Then by snake lemma ,the following sequence is exact

$$\ker \bar{f}_1 \to \ker f_2 \to \ker f_3|_{\operatorname{Im}\alpha_2} \to \operatorname{Coker} \bar{f}_1 \to \operatorname{Coker} f_2 \to \operatorname{Coker} f_3|_{\operatorname{Im}\alpha_2}$$

Furthermore, by applying the snake lemma to the following diagram

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Ker}\alpha_3 \arrow[r, "\iota_M"] \arrow[d, "f_3|_{\operatorname{Im}\alpha_2}"] & M_3 \arrow[r, "\alpha_3"] \arrow[d, "f_3"] & \operatorname{Im}\alpha_3 \arrow[r] \arrow[d, "f_4|_{\operatorname{Im}\alpha_3}"] & 0 \\
0 \arrow[r] & \operatorname{Ker}\beta_3 \arrow[r, "\iota_N"] & N_3 \arrow[r, "\beta_3"] & \operatorname{Im}\beta_3 \arrow[r] & 0
\end{tikzcd}$$

We get

$$0 = \ker f_4|_{\operatorname{Im}\alpha_3} \xrightarrow{\delta} \operatorname{Coker}\left(f_3|_{\operatorname{Im}\alpha_2}\right) \to \operatorname{Coker} f_3 = 0$$

which shows that $\operatorname{Coker}\left(f_3|_{\operatorname{Im}\alpha_2}\right) = 0$. Finally, we have the following sequence exact

$$0 = \operatorname{Coker} \bar{f}_1 \to \operatorname{Coker} f_2 \to \operatorname{Coker} f_3|_{\operatorname{Im}\alpha_2} = 0$$

Thus $\operatorname{Coker} f_2 = 0$, f_2 is surjective.

□

Corollary 0.4.4 ▸ Five lemma

In the following diagram of R-modules, the two rows are given to be exact. If f_1, f_2, f_4 and f_5 are isomorphisms then f_3 is also an isomorphism.

$$\begin{tikzcd}
M_1 \arrow[r] \arrow[d, "f_1"] & M_2 \arrow[r] \arrow[d, "f_2"] & M_3 \arrow[r] \arrow[d, "f_3"] & M_4 \arrow[r] \arrow[d, "f_4"] & M_5 \arrow[d, "f_5"] \\
M_1' \arrow[r] & M_2' \arrow[r] & M_3' \arrow[r] & M_4' \arrow[r] & M_5'
\end{tikzcd}$$

Proof. By applying the Four lemma to the left four columns, we get f_3 is injective. And by applying the Four lemma to the right four columns , we get f_3 is surjective .

□

Theorem 0.4.5

Given a short exact sequence of chain complexes

$$0 \to C'_* \xrightarrow{\alpha} C_* \xrightarrow{\beta} C''_* \to 0$$

there is a functorial long exact sequence of homology groups

$$\to H_n(C'_*) \xrightarrow{H_n(\alpha)} H_n(C_*) \xrightarrow{H_n(\beta)} H_n(C''_*) \xrightarrow{\delta_n} H_{n-1}(C'_*) \xrightarrow{H_{n-1}(\alpha)} H_{n-1}(C_*) \to$$

Proof sketch.
Consider the diagram

$$\begin{tikzcd}
 & C'_n/\mathrm{Im}\partial'_{n+1} \arrow[r, "\bar{\alpha}_n"] \arrow[d, "\partial'_n"] & C_n/\mathrm{Im}\partial_{n+1} \arrow[r, "\bar{\beta}_n"] \arrow[d, "\partial_n"] & C''_n/\mathrm{Im}\partial''_{n+1} \arrow[r] \arrow[d, "\partial''_{n+1}"] & 0 \\
0 \arrow[r] & \mathrm{Ker}\partial'_{n-1} \arrow[r, "\alpha'_{n-1}"] & \mathrm{Ker}\partial_{n-1} \arrow[r, "\beta'_{n-1}"] & \mathrm{Ker}\partial''_{n-1} &
\end{tikzcd}$$

Example 0.4.6 ▸ reduced homology

定义 ε 为将 0-链映为系数和的同态. 称同态 ε 为增广同态, 可以按以下方式扩张

出链复形 $\tilde{S}(X)$

$$\tilde{S}_n(X) = \begin{cases} S_n(X), & \text{if } n \geq 0, \\ \mathbb{Z}, & \text{if } n = -1, \\ (0), & \text{if } n < -1; \end{cases} \quad \text{and} \quad \tilde{\partial}_n = \begin{cases} \partial_n, & n \geq 1, \\ \varepsilon, & n = 0, \\ 0, & n < 0. \end{cases}$$

相应的同调群记作 $\tilde{H}_*(X)$，称为约化同调群. 显然 $\tilde{H}_i(X) = (0), i < 0; \tilde{H}_i(X) \simeq H_i(X), i > 1; \tilde{H}_0(X) \oplus \mathbb{Z} \simeq H_0(X)$. 特别地，对于道路连通空间 X, $\tilde{H}_0(X) = 0$.

Problem 0.4.7 ▸ cofibration

Let (X, A) be a pair of spaces. We say that the inclusion $A \to X$ is a cofibration if, whenever $f : X \to Y$ and $H : A \times [0,1] \to Y$ are maps such that $H(a,0) = f(a)$ for all $a \in A$, there exists $\tilde{H} : X \times [0,1] \to Y$ such that $\tilde{H}(a,t) = H(a,t)$ for all $(a,t) \in A \times [0,1]$ and $\tilde{H}(x,0) = f(x)$ for all $x \in X$.

1. Prove that the inclusion $(X, A) \to (X \cup CA, CA)$ induces an isomorphism on relative homology.
2. Prove that if $A \to X$ is a cofibration and A is contractible, then the quotient map $X \to X/A$ is a homotopy equivalence.
3. Prove that if X is obtained from A by gluing cells, then $A \to X$ is a cofibration.
4. Prove that if $A \to X$ is a cofibration, then $(X \cup CA, CA)$ is a cofibration.
5. Let $A \to X$ be a cofibration. Prove that the quotient map induces an isomorphism $H_i(X, A) \cong H_i(X/A, A/A) = \tilde{H}_i(X/A)$.

Problem 0.4.8 ▸ homology of suspension

Let X be a space. Compute $H_*(\Sigma X)$ in terms of $H_*(X)$.

Problem 0.4.9

Compute the homology of $\mathbb{CP}^n$.

0.5 Degree

Problem 0.5.1

Let $f : S^n \to S^n$ be a map. Its ***degree*** $\deg(f) \in \mathbb{Z}$ is the integer such that for all $x \in H_n(S^n; \mathbb{Z})$, we have $f_*(x) = \deg(f) \cdot x$.

1. Prove that this definition matches with the definition with fundamental groups for $n = 1$.
2. Prove that if $\deg(f) \neq 0$ then f is surjective. Find a counterexample for the converse.
3. Prove that if f is injective, then $\deg(f) = \pm 1$. Find a counterexample for the converse.
4. Prove that the degree of a reflection is -1 (use Mayer-Vietoris). Given $A \in O_n(\mathbb{R})$, prove that the degree of $x \mapsto Ax$ is equal to $\det(A)$. Compute the degree of the antipodal map $x \mapsto -x$.
5. Prove that ΣS^n is homeomorphic to S^{n+1}, then prove that the degree of f is equal to the degree of its suspension $\Sigma f : S^{n+1} \to S^{n+1}$.

Problem 0.5.2

1. Prove that any map $f : S^n \to S^n$ without fixpoints is homotopic to the antipodal map. (Use that for all $x \in S^n$, the origin $0 \in \mathbb{R}^{n+1}$ is not on the segment $[f(x), -x]$).
2. Prove that if n is even, then any $f : S^n \to S^n$ admits a fixpoint.
3. Prove that if a group G acts freely on S^n with n even, then $\#G \leq 2$.
4. Prove that S^n, for n odd, admits a nonvanishing vector field.
5. Prove that any vector field on S^n, for n even, vanishes at some point. (Given such a vector field X, consider the map $u \mapsto X(u)/\|X(u)\|$).